PRODUCT REQUIREMENTS DOCUMENT

Renewi — Contractor & Visitor Site Access Platform

Vlaardingen Lighthouse MVP → scalable NL & BE platform
Prepared from: Vlaardingen Contractor Access Discovery v1 (28 Aug 2026)
Implementation-oriented PRD for DigitalT3 / Kavia AI build

1. Executive Summary

Build a centralized digital Contractor & Visitor Site Access and Compliance platform for Renewi. The platform should manage company/person pre-registration, compliance evidence, training/tests, approval, access tokens, gate identity verification, on-site presence, dashboards, audit trails, and re-validation. The source deck describes six populations and NL/BE scope, with Vlaardingen proposed as the first lighthouse/MVP site. The source explicitly frames the requirements as a scope skeleton rather than a complete specification, so the build should be configurable and mock-first where Renewi-specific integrations or policy decisions are not yet available.

Source basis: the deck states that the ELT wants digital access and compliance control for everyone entering Renewi sites, with centralized compliance data and automated workflows; it proposes an MVP at one site designed to scale across NL and BE. filecite turn1file0 L93-L126

2. Product Vision

Create a single source of truth for who is authorized to enter Renewi sites, why they are entering, which company they represent, which compliance requirements they satisfy, when those requirements expire, and what happened at the gate.

- Prevent access by people who are not qualified, trained, identified, approved, or who are blacklisted.
- Reduce repeated administrative work through reusable person/company records and automated expiry/re-validation workflows.
- Make compliance evidence auditable and retrievable.
- Support very different arrival lead times, including short-notice external and agency drivers.
- Prove the thin slice at Vlaardingen while architecting for approximately 100 NL/BE sites.
- Keep the system usable even when upstream integrations, physical barrier hardware, or credentials are unavailable during development.

3. Source-of-Truth Scope

The source deck identifies six populations: Renewi drivers, temporary/agency drivers, external drivers, maintenance contractors, permanent contractors, and visitors. NL and BE are in scope; France and Portugal are explicitly out of scope for now. The deck estimates 1,500–2,500 people but marks this as low-confidence and says the number must be established rather than used as a firm sizing input. filecite turn1file0 L178-L217

Population	Definition	MVP priority
Renewi drivers	Own/permanent drivers; known and on HR record	Later
Temporary drivers	Agency-supplied; known to agency	MVP candidate
External drivers	Freight-exchange; may be unknown until arrival	MVP candidate
Maintenance contractors	Ad hoc; short notice	Later
Permanent contractors	Fixed-term; longer lead time	Later
Visitors	Hosted; lightest check set	Later

4. Compliance Model

The source check matrix covers corporate compliance, personal identification, driving licence, VCA, safety training, site rules, access approval token, and blacklist. Visitors have the lightest check set. Blank matrix cells are explicitly identified in the deck as an inference to confirm, so the implementation must make rule applicability configurable rather than hard-code assumptions. filecite turn1file0 L221-L248

Requirement	Drivers	Contractors	Visitors	System behavior
Corporate compliance	Yes	Yes	No	Company compliance status + evidence + expiry
Personal identification	Yes	Yes	Yes	Registered identity evidence; gate verification
Licence to drive	Yes	No	No	Licence type/status/expiry
VCA	No	Yes	No	Certificate/status/expiry
Safety training	Yes	Yes	Yes	Training completion + score
Site rules	Yes	Yes	Yes	Site-specific module + test + score
On-site access approved	Yes	Yes	Yes	Approval state + QR/badge token
Blacklist	Yes	Yes	Yes	Person/company block with reason and audit history

5. End-to-End Target Journey

1. Invite/request: planner, host, or authorized requester initiates access.
2. Pre-register: company first, then person and visit/site.
3. Capture evidence: upload or capture ID, licence, VCA, insurance and other configured evidence.
4. Train & test: assign relevant SHEQ and site-rule modules; record completion and score.
5. Approve: evaluate required checks, validity, blacklist status and authorization.
6. Issue access token: generate QR token; support future badge integration.
7. Arrive at gate: identify the person and retrieve the active visit/authorization.
8. Grant or deny: show a clear decision and exception path.
9. On-site record: record entry, current presence and exit.
10. Dashboard: show daily/site presence, compliance, pending approvals, expiring items and audit trail.
11. Re-validation: notify before expiry, re-check evidence/training and update authorization.

This eleven-step journey is a DigitalT3 reconstruction/hypothesis from the deck, not an observed current-state process. The PRD therefore treats it as the target-state baseline to validate. filecite turn1file0 L255-L320

6. MVP Definition — Vlaardingen

The source deck proposes a candidate MVP centered on external and agency drivers because they are the populations with least current visibility. The proposed slice is self-contained and exercises enrolment, evidence, training, token, gate verification and dashboard. filecite turn1file0 L542-L602

- Site: Vlaardingen (configurable; treated as proposed pilot).
- Populations: external drivers and temporary/agency drivers.
- Manual enrolment is acceptable for MVP; deep HR/agency integrations are not required to prove the flow.
- QR access token is the primary MVP token.
- Physical badge hardware and barrier integration are out of the first slice.

- France/Portugal and Seveso-specific regime are out of the first slice.
- Architecture must still support later population, site, integration and country expansion.

7. User Roles & Permissions

Role	Primary capabilities
Platform Admin	Global configuration, users/roles, sites, rule templates, audit, system health
Renewi Compliance/SHEQ	Compliance rules, evidence review, training/tests, exceptions, blacklist governance
Site Admin / Gate Manager	Site configuration, daily arrivals, gate operations, deny/approve overrides within policy
Planner / Requester	Invite/register companies/persons/visits; monitor status
Contractor/Agency User	Register persons, upload evidence, track compliance and visits
Driver/Visitor	Complete registration, upload/confirm evidence, training/tests, present QR
Gate Operator	Scan QR, identify/verify, view pass/fail decision, record entry/exit, handle exceptions
Auditor / Read-only	Search evidence, decisions, access events and audit history

8. Functional Requirements

8.1 Organization, Company & Person Registry

- Create/search/update company records with legal identity, supplier status, insurance/compliance status and documents.
- Create/search/update person records with population type, contact details, company relationship, status and registration history.
- Maintain one reusable person record and one reusable company record, with site-specific authorization layered on top.
- Support duplicate detection using configurable identity/company matching fields.
- Record source of registration: Renewi, agency, freight exchange, contractor, visitor host, manual.

8.2 Visit / Access Request

- Create a visit request linked to site, company, person, purpose, requester/host, planned arrival/departure and population.
- Support one-off and repeat visits.
- Support short-notice registration with a fast path and explicit exception handling.
- Display a completion/readiness score for the visit.

8.3 Evidence & Document Intelligence

- Upload documents through drag-and-drop and mobile-friendly capture.
- Extract structured fields from supported evidence using OCR/document intelligence.
- Store original evidence plus extracted fields and verification status.
- Track document type, issuer, issue date, expiry date, document number and confidence.
- Flag low-confidence extraction for human review.
- Never silently mark a document compliant solely from OCR; approval requires configured verification rules.
- Design for ID, driving licence, VCA and insurance evidence.

8.4 Compliance Rules Engine

- Configurable rules by population, site, country and requirement.

- Each rule defines evidence required, verification method, validity period, re-validation trigger, failure behavior and approver.
- Support AND/OR rule composition where needed.
- Calculate overall compliance as Pass / Fail / Pending / Expired / Blocked / Exception.
- Expose exactly which checks caused a failed or pending decision.

8.5 Training & Testing

- Create configurable training modules and site-rule modules.
- Assign modules based on population/site/rule.
- Record content version, completion timestamp, score, pass threshold and attempt history.
- Support multilingual content architecture; do not hard-code Dutch-only UX.
- Do not author Renewi SHEQ content as part of the product; provide content management/import hooks.

8.6 Blacklist / Blocklist

- Blacklist person and/or company.
- Require reason, effective date, optional expiry and actor.
- Apply blacklist to gate decision immediately.
- Support controlled override only for explicitly authorized roles and require justification.
- Maintain immutable audit events for additions, changes and overrides.

8.7 Approval & QR Access Token

- Generate a cryptographically random, short-lived visit/access token represented as QR.
- QR must not expose sensitive personal data.
- Token states: issued, active, used, expired, revoked, denied.
- Allow re-issue/revocation.
- Keep token abstraction separate from future badge/physical access integrations.

8.8 Gate Operations

- Dedicated gate screen optimized for speed and large touch targets.
- Scan QR or search by authorized fallback identifier.
- Retrieve current visit and compliance decision.
- Show clear Grant / Deny result with reason categories.
- Record gate event, operator/device, timestamp, site, token and decision.
- Support entry and exit events.
- Provide offline-capable architecture with a controlled cached-read/fallback mode; exact offline policy remains configurable pending discovery.

8.9 Identity Verification

- Design an identity-verification adapter that can support document/photo comparison and later approved live identity matching.
- MVP should include a simulated/mock verification provider so the complete journey works without third-party credentials.
- Do not claim biometric/legal compliance in the UI; label this as configurable/pending Renewi privacy approval.
- Keep biometric processing isolated behind an adapter boundary.

8.10 Dashboard & Reporting

- Site/day dashboard: expected, arrived, currently on site, exited, denied.
- Compliance dashboard: compliant, pending, expired, blocked, expiring soon.
- Registration pipeline: invitations, evidence pending, training pending, approval pending.
- Exceptions: denied entries, overrides, failed checks.

- Search/filter by site, date, population, company, person, status and requirement.
- Export audit/compliance reports where permitted.

8.11 Notifications & Re-validation

- Expiry reminders at configurable intervals.
- Notify responsible parties about missing/expired evidence or training.
- Escalate unresolved expiries.
- Recompute authorization whenever a required check changes.
- Keep notification delivery behind an adapter so email/SMS/enterprise messaging can be connected later.

8.12 Audit & Evidence

- Immutable audit log for compliance state changes, approvals, denials, blacklist events, document changes, gate events and overrides.
- Audit entries include actor, role, timestamp, action, entity, previous/new state and reason where relevant.
- Evidence access itself should be auditable.
- Provide traceability from a gate decision back to each rule/evidence item.

9. Non-Functional Requirements

Area	Requirement
Security	RBAC, secure sessions, least privilege, server-side authorization, input validation, encrypted transport, secrets via environment variables
Privacy	Data minimization, configurable retention, access logging, deletion/anonymization hooks, privacy-by-design; DPIA/legal decisions remain external dependencies
Performance	Normal admin pages should feel responsive; gate decision path should be optimized for low latency and minimal clicks
Availability	Production architecture should support resilient deployment; MVP should have graceful error/fallback states
Scalability	Multi-site, multi-country, multi-population, configurable rules and adapters from day one
Accessibility	Keyboard support, readable contrast, clear status semantics, large gate controls, WCAG-oriented implementation
Localization	i18n-ready; avoid Dutch-only hard-coded strings
Observability	Structured logs, health endpoints, error tracking hooks, metrics-ready architecture
Auditability	Every compliance decision must be explainable from stored rule/evidence state
Maintainability	Modular services, typed APIs, migrations, tests, linting and clear README

10. Recommended Technology Stack

This stack is a DigitalT3 implementation recommendation, not a technology choice stated by the source deck. It is optimized for a Kavia-generated codebase that can be downloaded and then maintained locally in VS Code.

Layer	Recommendation	Reason
Frontend	Next.js 15+ (App Router) + TypeScript	Production React architecture, routing, server/client components
UI	Tailwind CSS + shadcn/ui + Radix primitives	Fast, consistent, professional enterprise UI
Data fetching	TanStack Query	Server-state caching, mutations and

		resilient async UX
Forms/validation	React Hook Form + Zod	Typed validation and scalable forms
Charts	Recharts	Dashboard analytics without heavy dependency
Backend	Python 3.12 + FastAPI	Strong API ergonomics; suitable for document/AI integrations
ORM	SQLAlchemy 2.x + Alembic	Explicit relational model and migrations
Database	PostgreSQL 16+	Relational integrity, auditability and scale
Object storage	S3-compatible storage; local filesystem adapter for dev	Evidence/document storage with provider abstraction
Auth	JWT/session-based RBAC; provider adapter	Credential-free mock mode first; enterprise IdP can be added later
AI/OCR	Provider-agnostic document intelligence adapter; local/mock extractor for MVP	Avoid credential blockers while preserving production integration boundary
Identity	Provider-agnostic verification adapter; mock implementation for MVP	Separates sensitive identity/biometric integration from core workflow
QR	Standard QR encoder/decoder library	Simple MVP access-token mechanism
Background jobs	Redis + Celery/ARQ-compatible worker layer	Expiry reminders, OCR jobs and asynchronous processing
API docs	OpenAPI generated by FastAPI	Easy integration and debugging
Testing	Pytest + Playwright + Vitest/React Testing Library	Backend, E2E and UI coverage
DevOps	Docker + Docker Compose	Reproducible local environment
CI	GitHub Actions	Lint, type-check, test and build
Deployment	Railway-compatible Docker deployment	Matches the user's preferred post-Kavia local/deployment workflow
Observability	Structured JSON logging + Sentry adapter + health/readiness endpoints	Production diagnostics without hard dependency

11. High-Level Architecture

Use a modular monolith for MVP rather than premature microservices. Keep boundaries clean enough that document processing, identity verification, notifications and enterprise integrations can later become separate services.

- Web App → FastAPI API → domain/application services → PostgreSQL.
- Evidence uploads → object-storage adapter → document-processing worker → extracted metadata → compliance engine.
- Gate UI → QR/token verification API → compliance engine → gate event/audit log.
- Training UI → training service → attempts/scores → compliance engine.
- Scheduler/worker → expiry detection → notification adapter → re-validation workflow.
- Integration adapters: HR, agency, LMS/SHEQ, access control, identity provider, freight-exchange sources.
- All external providers must have interfaces plus mock/local implementations.

12. Core Data Model

Entity	Key fields
User	id, role, name, email, status, auth provider
Company	id, legal name, registration identifiers, type, compliance status
Person	id, company_id, population, identity metadata, status
Site	id, name, country, address, timezone, gate configuration, active
RequirementRule	id, site/population scope, requirement type, mandatory,

	validity, verification policy
Evidence	id, person/company, type, file reference, extracted fields, confidence, status, expiry
TrainingModule	id, type, site, locale, version, pass threshold, active
TrainingAttempt	id, person, module, score, result, version, timestamps
Visit	id, site, person, requester/host, purpose, planned times, status
ComplianceCheck	id, visit/person, rule, status, evidence reference, evaluated_at
AccessAuthorization	id, visit, status, token, valid_from/to
GateEvent	id, site, visit/person, event_type, decision, timestamp, operator/device
BlacklistEntry	id, person/company, reason, effective/expiry, status
Notification	id, recipient, type, status, related entity, scheduled/sent time
AuditEvent	id, actor, action, entity, before/after, reason, timestamp
IntegrationConnection	id, provider, status, configuration metadata, last sync

13. UI/UX Requirements

The UI must be fully interactive, professional and production-level. Do not build a collection of static CRUD screens.

- Enterprise operations aesthetic: clean information hierarchy, restrained visual language, dense but readable tables, clear status chips and strong empty/error/loading states.
- Responsive layouts for desktop admin users and tablet/mobile gate operations.
- Persistent navigation with role-aware modules.
- Global search/command affordance for quickly finding people, companies, visits and sites.
- Interactive dashboards with drill-down from KPI → list → record → evidence/audit.
- Step-based registration wizard with progress indicator and save/resume behavior.
- Compliance readiness visualization that clearly shows why someone is not ready.
- Document viewer with extracted-field side panel and verification controls.
- Gate mode should be visually distinct, simplified and optimized for speed.
- Use confirmations for consequential actions such as blacklist, deny, revoke and override.
- Provide skeleton loaders, optimistic updates where safe, toast feedback, inline validation and useful error recovery.
- Avoid excessive modal usage; use drawers/detail pages for complex workflows.
- Use consistent icons, keyboard navigation, focus states and accessible semantics.
- No placeholder-looking UI in final MVP.

14. Key Screens

12. Login / demo access
13. Global operations dashboard
14. Site dashboard
15. People directory
16. Company directory
17. Person profile + compliance timeline
18. Company profile + compliance
19. New access request wizard
20. Registration status / checklist
21. Evidence upload & verification
22. Training & site-rules learning screen
23. Compliance decision center
24. Blacklist management

25. QR pass / access token
26. Gate operator screen
27. Live on-site roster
28. Visit detail
29. Audit trail
30. Expiry / re-validation center
31. System settings / rules / sites / integrations
32. Requirements & credential readiness page

15. Demo / Mock-First Strategy

The user explicitly wants Kavia to finish the project without stopping for credentials. Therefore the repository must contain a complete runnable demo mode using seeded deterministic data and mock adapters.

- Seed Vlaardingen site, sample companies, people from all six populations, sample evidence, training modules, visits, QR passes and gate events.
- Include at least one fully compliant person, one pending evidence case, one expired document case, one failed training case, one blacklisted person and one denied gate case.
- Mock OCR returns deterministic extracted fields and confidence scores.
- Mock identity verification returns configurable match/no-match outcomes.
- Mock notification provider logs deliveries in the database/application logs.
- Mock HR/agency/LMS/access-control connectors expose realistic integration interfaces but do not require credentials.
- Demo login accounts for each role must be documented without using real secrets.
- Provide reset/seed scripts so the demo can be restored quickly.

16. Requirements & Credentials Documentation

Create a /requirements directory in the repository. It is a first-class deliverable, not a placeholder.

- requirements/README.md — project setup, local run, architecture and completion checklist.
- requirements/credentials.md — every external credential required for production, provider, environment variable name, where to obtain it, why it is needed, scope/permissions, and whether optional.
- requirements/integrations.md — HR, agency, LMS/SHEQ, access control, identity verification, notification and storage integration contracts.
- requirements/privacy.md — data categories, retention placeholders, DPIA/legal questions and decisions still required from Renewi.
- requirements/site-readiness.md — gate/network/device/offline assumptions and per-site readiness fields.
- requirements/demo.md — demo users, seeded scenarios and reset instructions.
- requirements/deployment.md — environment variables, Docker/Railway deployment and production checklist.
- requirements/open-items.md — unresolved source-deck questions and implementation assumptions, clearly labelled.

The source deck specifically identifies unresolved questions around incumbent capabilities, authoritative records, gate reality, DPIA, training systems, IT standards and volumes. These should be represented as open items rather than silently invented requirements. filecite turn1file0 L404-L417

17. Security & Privacy Guardrails

- Never commit secrets, tokens, certificates or production credentials.
- Use .env.example with variable names only.
- Separate PII/identity evidence from general application metadata where practical.
- Restrict access to ID documents and identity-related information by role.

- Log access to sensitive evidence.
- Use signed/opaque access tokens rather than embedding PII in QR payloads.
- Implement configurable retention/deletion hooks.
- Treat live identity matching/biometric functionality as gated by Renewi privacy/legal approval; provide mock functionality for development.
- Do not claim a DPIA is complete; the source deck says this remains an open question. filecite turn1file0 L382-L396

18. API Surface — Initial

- Auth: login, session, current user.
- People: CRUD, search, compliance summary, history.
- Companies: CRUD, compliance, evidence.
- Sites: CRUD, site rules, gate configuration.
- Visits: create, update, invite, status, check-in/out.
- Evidence: upload, extract, review, verify, expire.
- Training: modules, assignments, attempts, results.
- Compliance: evaluate, explain, approve, reject, override.
- Blacklist: add, update, remove/expire, check.
- Access: issue/revoke/validate QR tokens.
- Gate: scan/verify, decision, entry, exit.
- Dashboard: KPIs, roster, trends, exceptions.
- Audit: search/filter/export.
- Notifications: schedule, send, delivery history.
- Integrations: connector status, sync, test connection (mock-first).

19. Implementation Plan — Bunch-by-Bunch

Bunch	Deliverables	Exit criteria
0 — Foundation	Repo structure, requirements docs, architecture, Docker, env examples, lint/type/test baseline	Fresh clone starts locally; requirements directory exists
1 — Data & API foundation	PostgreSQL schema, migrations, seed/reset, FastAPI modules, RBAC skeleton	Health endpoint + migrations + seeded data work
2 — Core registry	People, companies, sites, visits, search/filter/detail pages	CRUD and navigation work end-to-end
3 — Compliance engine	Rule model, applicability, statuses, evaluation/explanation	Sample scenarios produce deterministic decisions
4 — Evidence intelligence	Upload, storage adapter, OCR mock, extracted fields, review UI	Evidence flows from upload to verified/failed state
5 — Training	Modules, assignments, tests, scoring, site rules	Training result affects compliance
6 — Approval & QR	Approval workflow, token issuance/revocation, QR UI	Approved visit gets valid QR; denied/expired token fails
7 — Gate operations	Gate screen, QR scan, decision, entry/exit, roster	Complete simulated arrival flow works
8 — Dashboard & audit	Operations KPIs, on-site roster, exceptions, audit trail	Decision is traceable and visible in dashboard
9 — Re-validation	Expiry engine, notifications mock, renewal workflow	Expiring evidence causes predictable alerts/status changes
10 — Production hardening	Security, accessibility, errors, tests, performance, observability, docs	Quality gates pass; no known blocker
11 — Finalization	Demo seed, deployment configs,	Project is runnable without external

	requirements/credentials completion, final README	credentials and production dependencies are documented

20. Acceptance Criteria

- A new external driver can be registered from invitation through evidence, training, approval and QR issuance using only local/mock services.
- A gate operator can scan/enter the QR, receive an explainable grant/deny decision and record entry.
- A denied case clearly identifies the failed requirement.
- A blacklisted person cannot receive a successful gate authorization.
- An expired licence/VCA/insurance/training requirement causes the configured compliance state to update.
- A site-specific rule can make an otherwise compliant person non-compliant.
- A visitor follows a lighter configurable requirement path.
- A person/company can be reused across visits without re-entering core data.
- All consequential changes are visible in the audit trail.
- Dashboard accurately reflects current on-site state from gate events.
- The system runs with zero real third-party credentials in demo mode.
- All production credentials and external dependencies are listed in requirements/credentials.md at project completion.
- Docker Compose can start the full local stack from a clean checkout.
- Database migrations and deterministic seed/reset scripts work.
- Frontend/backend tests and builds complete successfully.

21. Explicit Out of Scope / Not to Invent

- Do not implement physical barrier hardware integration in the MVP.
- Do not require deep HR or agency integrations for the MVP.
- Do not build France or Portugal support into the first delivery beyond extensible country configuration.
- Do not implement a Seveso-specific regime unless explicitly reintroduced after discovery.
- Do not author Renewi's SHEQ content as product content.
- Do not assume the dashboard is a real-time emergency muster system; the source deck explicitly flags this as an assumption to resolve.
- Do not invent legal basis, retention periods, DPIA approval, biometric processing policy or labour/works-council decisions.
- Do not invent real vendor names, credentials, endpoints or enterprise systems.

22. Open Questions Carried Forward

- Does an existing Renewi contractor/site-access capability already exist, especially at Seveso sites?
- What is the authoritative record for a person and company?
- Which population is formally selected for MVP?
- What happens today when a check fails, and who decides?
- Is a DPIA required for ID capture/live identity matching?
- What is the physical gate reality at Vlaardingen?
- Where do SHEQ training and tests currently live?
- Which IT platforms/standards/environments are mandatory?
- What are actual volumes and repeat-visit ratios?

These are directly carried from the source deck's gating questions. filecite turn1file0 L404-L417

23. Definition of Done

- All implementation bunches completed, not merely scaffolded.
- No TODOs or placeholder pages in core user journeys.
- Core flows demonstrated using seeded data.
- All services documented and runnable locally.
- Credential-free mock mode works from clean checkout.
- Production adapters/interfaces exist where external credentials would eventually be required.
- Requirements directory is complete and lists every credential/configuration dependency discovered during implementation.
- README includes exact local setup, demo accounts, seed/reset, test and deployment commands.
- Final project can be downloaded from Kavia and continued in VS Code without needing Kavia for routine development.

24. Source / Interpretation Note

The uploaded deck is a discovery proposal, not a finalized technical specification. It explicitly distinguishes what is stated in the original requirements sheet from DigitalT3 inference and says the candidate MVP, journey, DPIA assumption and discovery shape are hypotheses to validate. This PRD therefore converts the stated scope into buildable product requirements while keeping unresolved items configurable and documented. filecite turn1file0 L692-L724

Prepared for DigitalT3 implementation planning. This document does not replace Renewi's legal, privacy, IT architecture or operational approvals.